

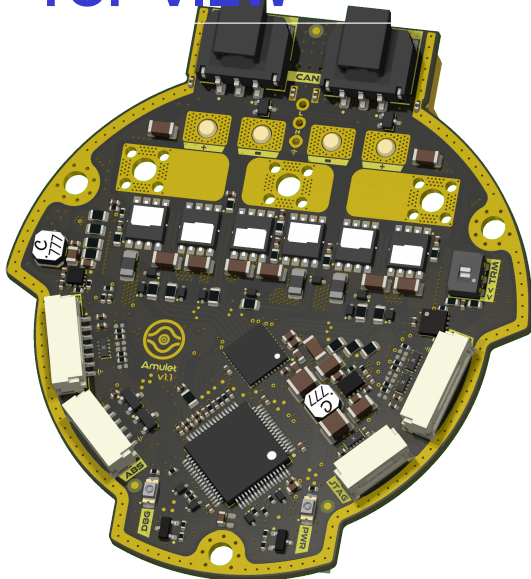
Amulet Motion Controller

Variant: Released

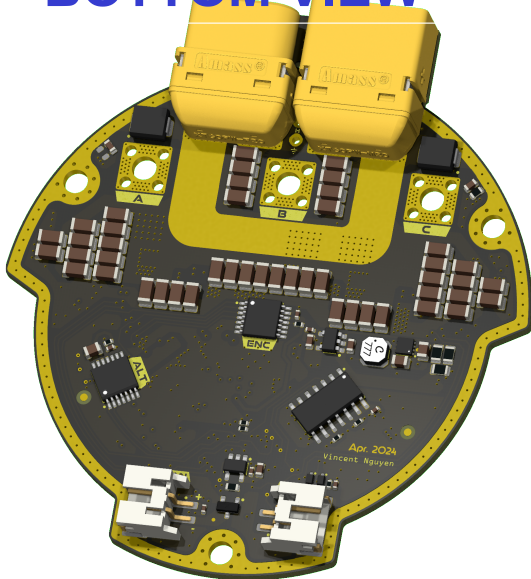
2024-11-13
Rev 1.1

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TOP VIEW



BOTTOM VIEW



NOTES

Schematic based off Josh Pieper's moteus controllers.

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Released 13-APR-2024

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for
informational design
notes.

DESIGN NOTE:
Example text for
debug notes.

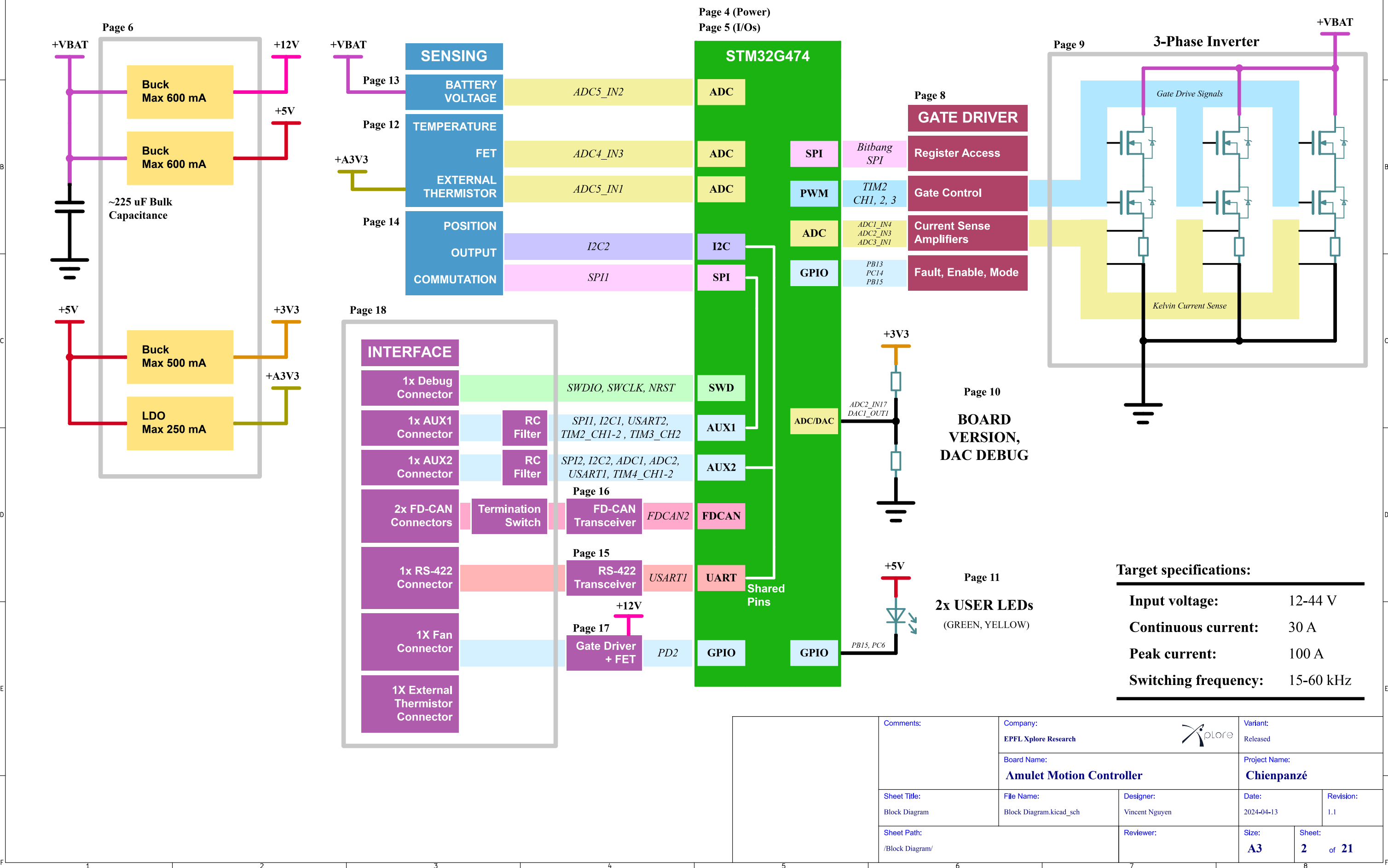
DESIGN NOTE:
Example text for
cautionary design
notes.

DESIGN NOTE:
Example text for
critical design
notes.

LAYOUT NOTE:
Example text for
critical layout
guidelines.

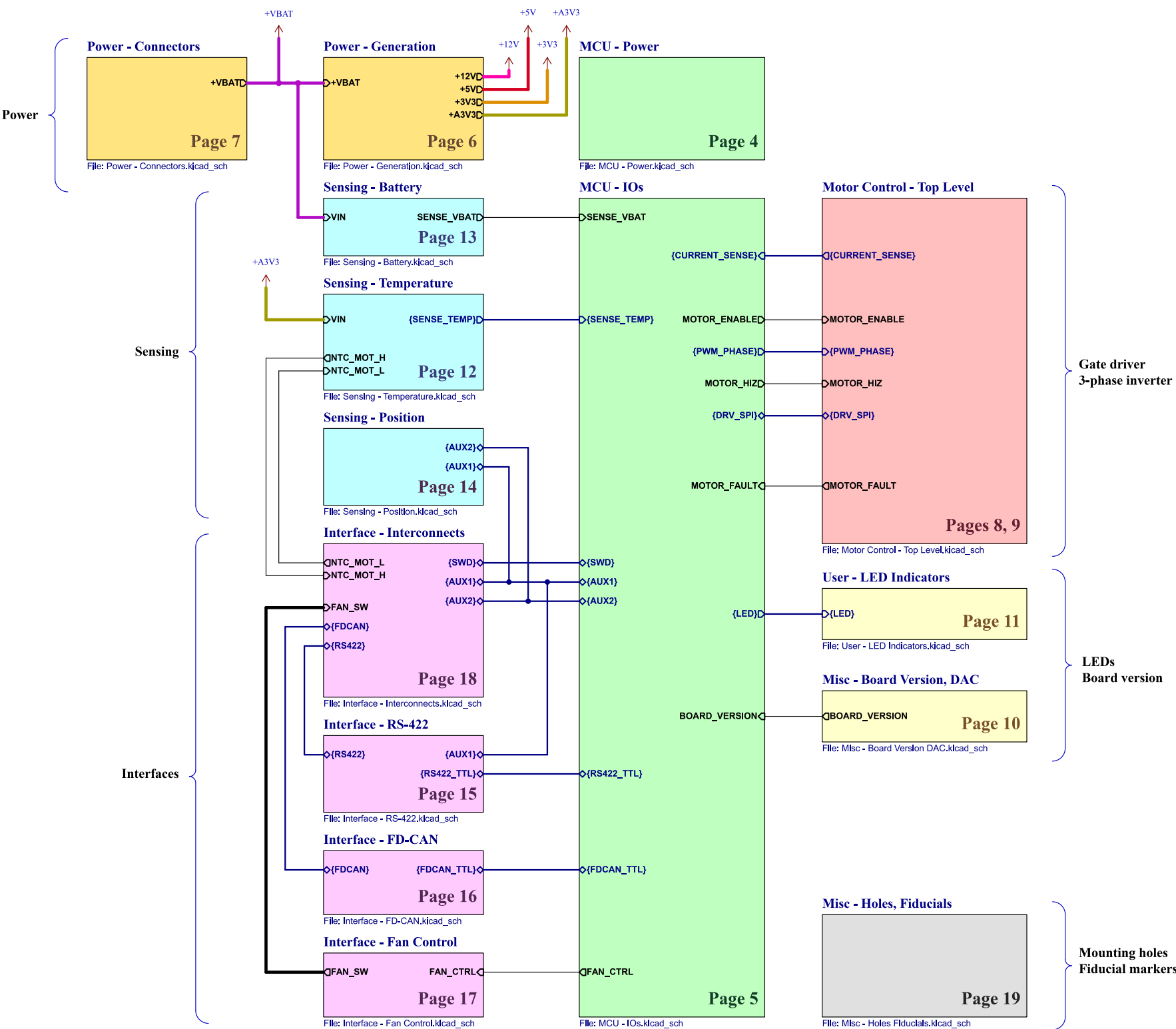
	Comments:	Company: EPFL Xplore Research	Variant: Released	
	Sheet Title: Cover Page	Board Name: Amulet Motion Controller	Project Name: Chienpanzé	
	Sheet Path: /	File Name: amulet_controller.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13
			Reviewer:	Revision: 1.1
			Size: A3	Sheet: 1 of 21

[2] Block Diagram



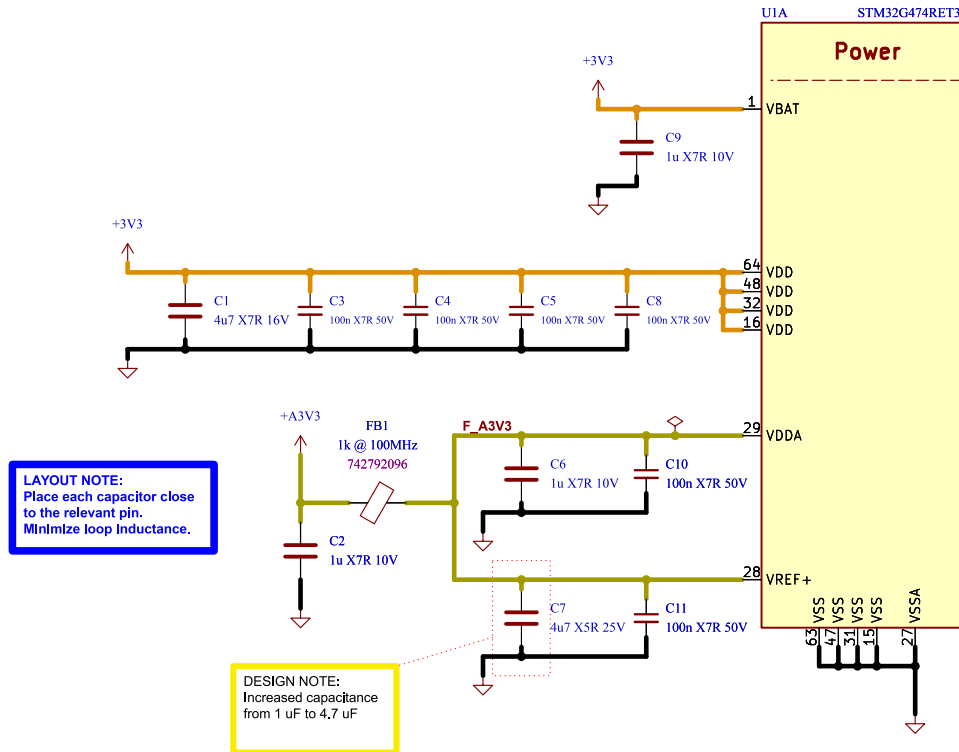
Comments:	Company: EPFL Xplore Research				Variant: Released	
	Board Name: Amulet Motion Controller			Project Name: Chienpanzé		
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13		Revision: 1.1	
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 21		

[3] Project Architecture



Comments:	Company: EPFL Xplore Research				Variant: Released	
	Board Name: Amulet Motion Controller			Project Name: Chienpanzé		
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-22	Revision: 1.1		
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 21		

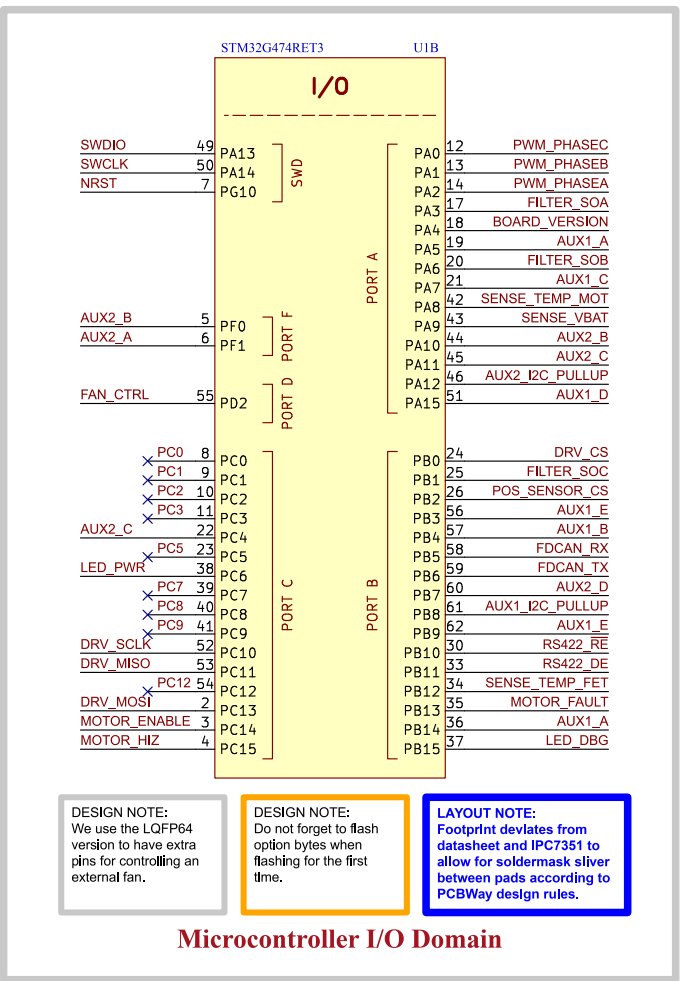
[4] MCU - Power



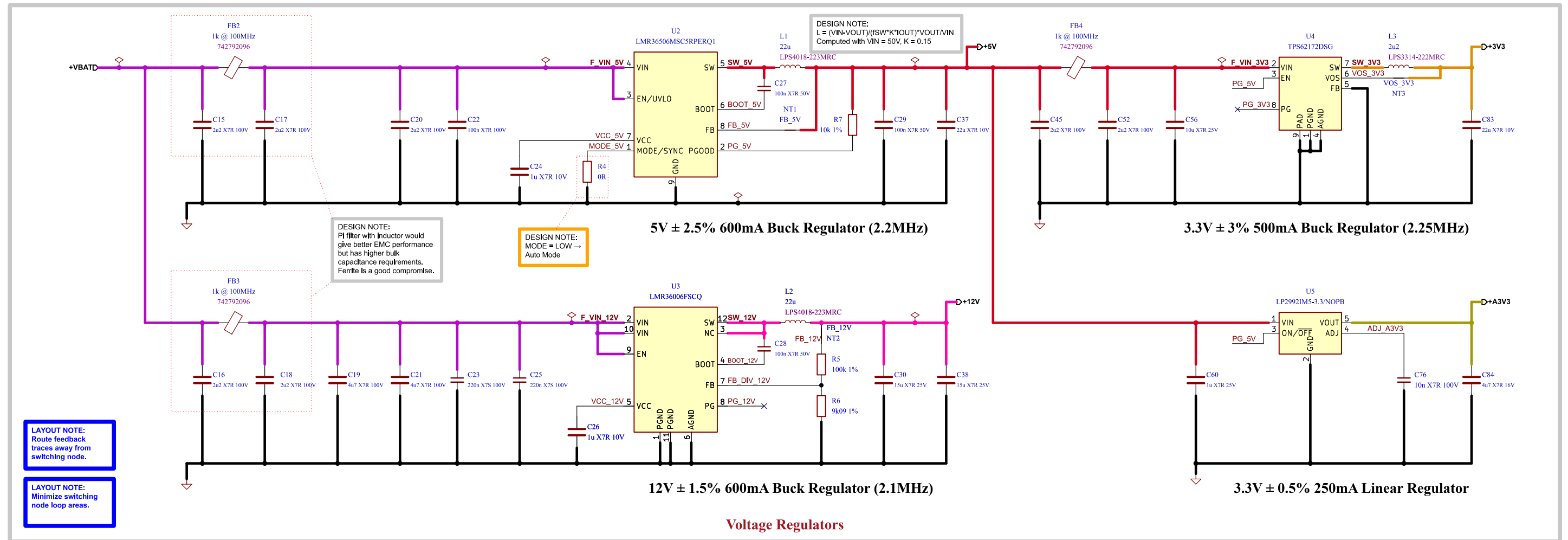
Microcontroller Power Supply Scheme

	Comments: AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation		Company: EPFL Xplore Research		 Variant: Released		
			Board Name: Amulet Motion Controller		Project Name: Chienpanzé		
	Sheet Title: MCU - Power		File Name: MCU - Power.kicad_sch		Designer: Vincent Nguyen		
				Date: 2023-12-18		Revision: 1.1	
Sheet Path: /Project Architecture/MCU - Power/				Reviewer:		Size: A4	
						Sheet: 4 of 21	

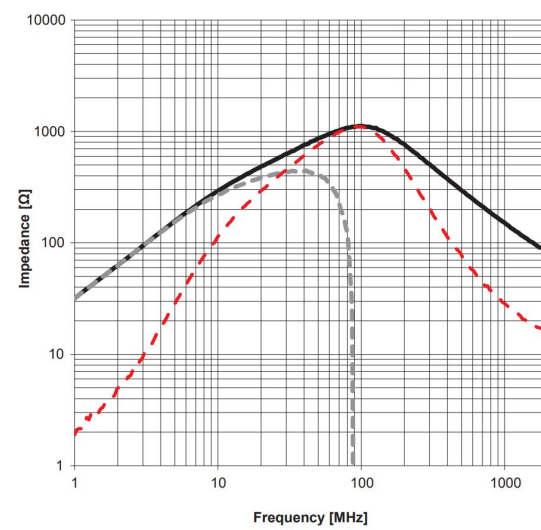
[5] MCU - I/Os



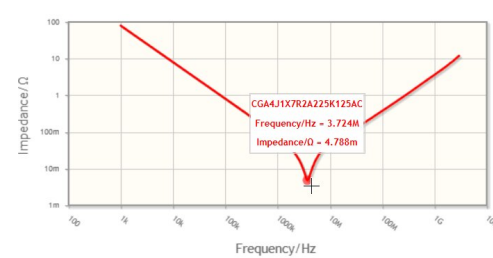
[6] Power - Generation



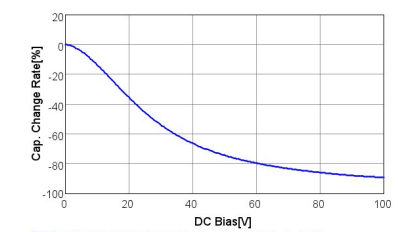
Ferrite bead Impedance vs Frequency curve



Pi filter capacitor Impedance vs Frequency curve



Bulk capacitance Derating curve



Capacitor and Ferrite Bead Parameters

Battery

+VBAT TP39 +VBAT
+VBAT TP40 +VBAT
+VBAT TP41 +VBAT

Filtered Power Rails

F_VIN_12V TP42 F_VIN_12V
F_VIN_5V TP43 F_VIN_5V
F_VIN_3V3 TP44 F_VIN_3V3

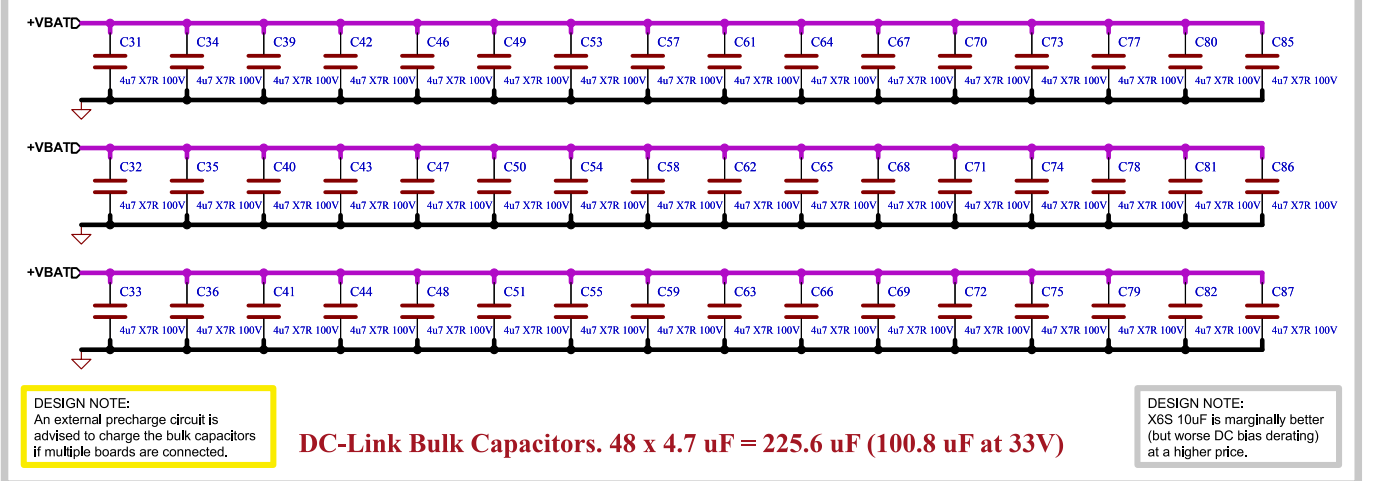
Power Rails

+12V TP45 +12V
+5V TP46 +5V
+5V TP47 +5V
+3V3 TP48 +3V3
+3V3 TP49 +3V3
+3V3 TP50 +3V3
+A3V3 TP51 +A3V3

Ground

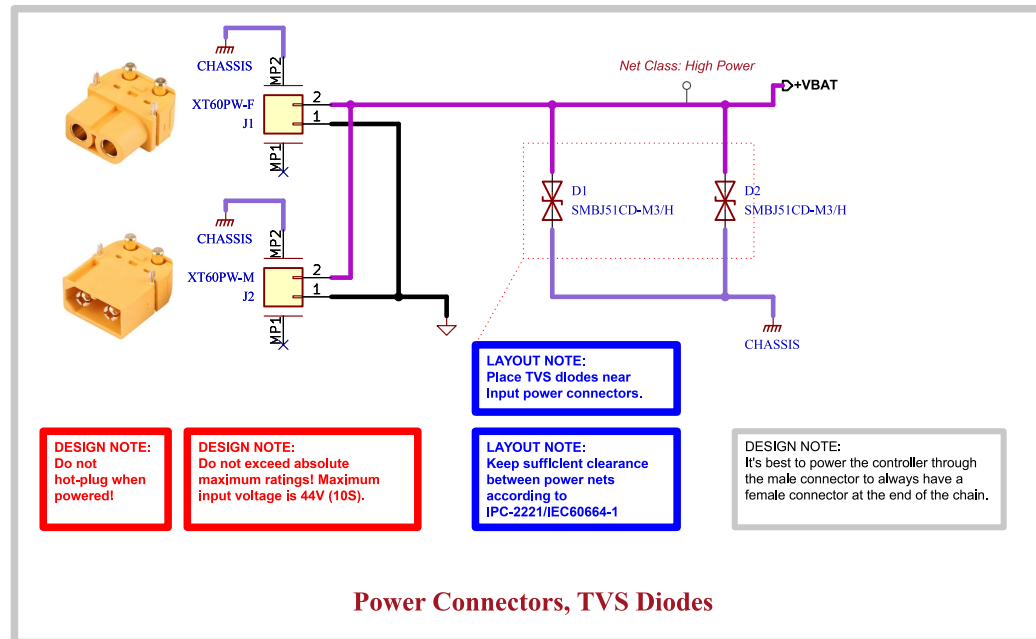
GND TP52 GND
GND TP53 GND
GND TP54 GND
GND TP55 GND
GND TP56 GND
GND TP57 GND
GND TP58 GND

Test Points



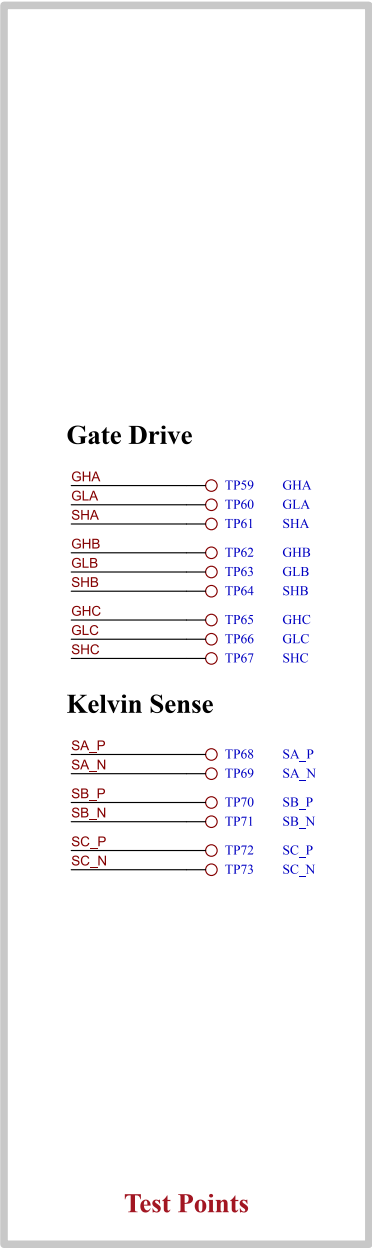
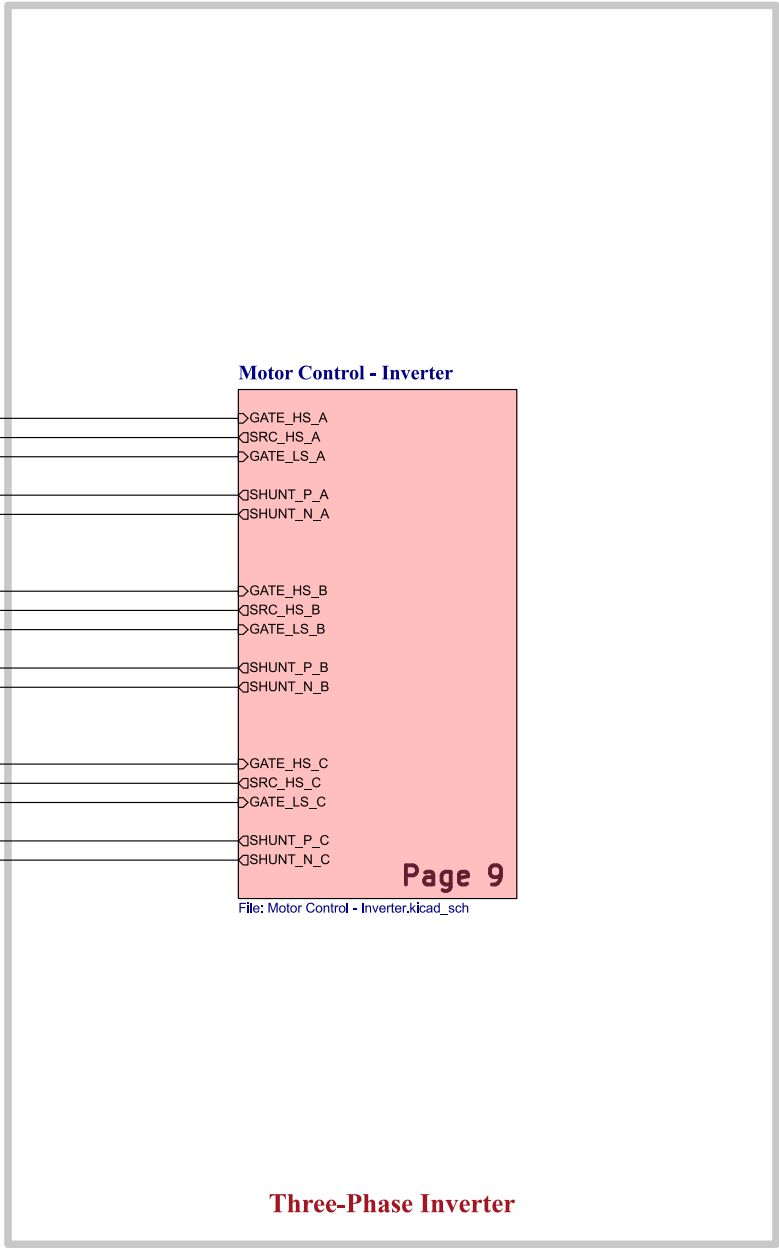
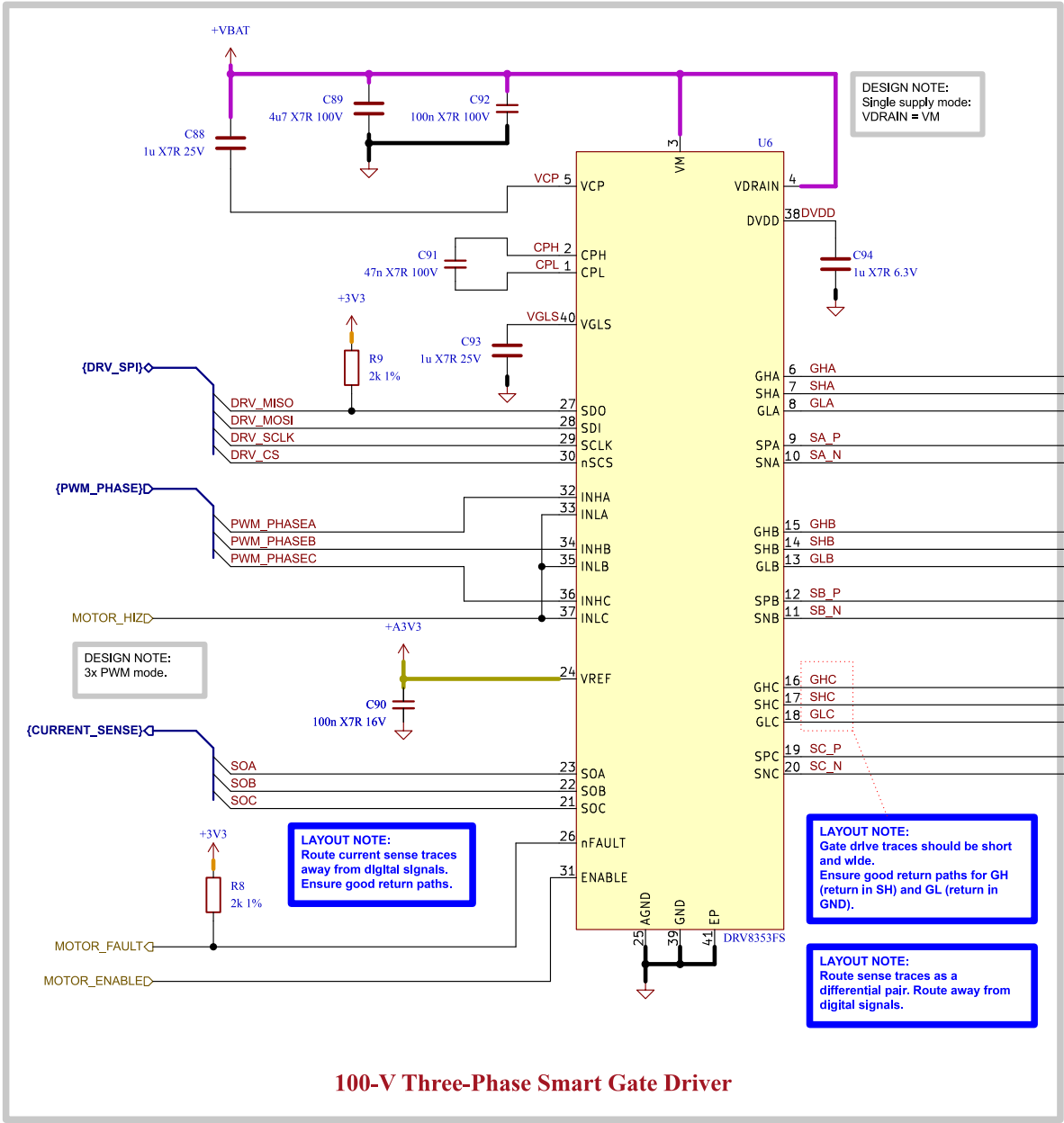
Comments: +12V (600 mA): Fan power +5V (600 mA): Gate Driver, LEDs +3V3 (500 mA): digital 3.3V +3V3A (250 mA): analog 3.3V	Company: EPFL Xplore Research		Variant: Released	
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Power - Generation	File Name: Power - Generation.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13
Sheet Path: /Project Architecture/Power - Generation/	Reviewer:	Size: A3	Sheet: 6 of 21	Revision: 1.1

[7] Power - Connectors



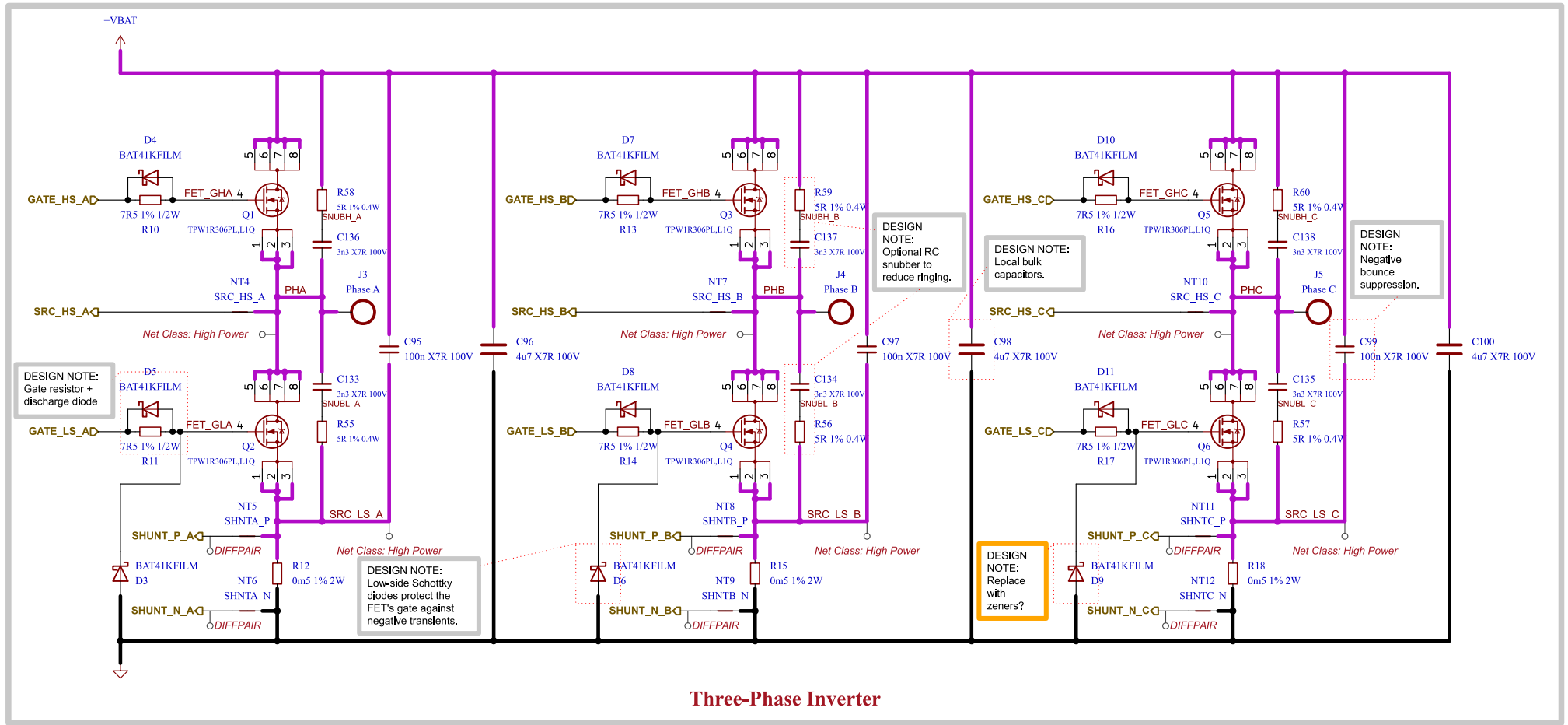
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	Sheet Title: Power - Connectors	File Name: Power - Connectors.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31
	Sheet Path: /Project Architecture/Power - Connectors/	Reviewer:	Size: A4	Revision: 1.1
			Sheet: 7	of 21

[8] Motor Control - Top Level



	Comments:		Company:				Variant:		
			EPFL Xplore Research				Released		
			Board Name:		Project Name:				
			Amulet Motion Controller		Chienpanzé				
	Sheet Title:		File Name:		Designer:		Date:		Revision:
Motor Control - Top Level		Motor Control - Top Level.kicad_sch		Vincent Nguyen		2023-12-20		1.1	
Sheet Path:				Reviewer:		Size:		Sheet:	
/Project Architecture/Motor Control - Top Level/						A3		8 of 21	

[9] Motor Control - Inverter



LAYOUT NOTE:
High current traces must be carefully designed. Ensure ground return path does not cross sensitive parts of the board. Use multiple planes for higher current carrying capacity.

LAYOUT NOTE:
Keep sufficient clearance between power nets according to IPC-2221/IEC60664-1.

DESIGN NOTE:
A gate drive current that is too large can damage the FETs!

Comments:
System Design Considerations for High-Power Motor Driver Applications
Best Practices for Board Layout of Motor Drivers
Proper RC Snubber Design for Motor Drivers

Sheet Title:
Motor Control - Inverter

Sheet Path:
/Project Architecture/Motor Control - Top Level/Motor Control - Inverter/

Company:
EPFL Xplore Research

Board Name:
Amulet Motion Controller

File Name:
Motor Control - Inverter.kicad_sch

Reviewer:



Variant:
Released

Project Name:
Chienpanzé

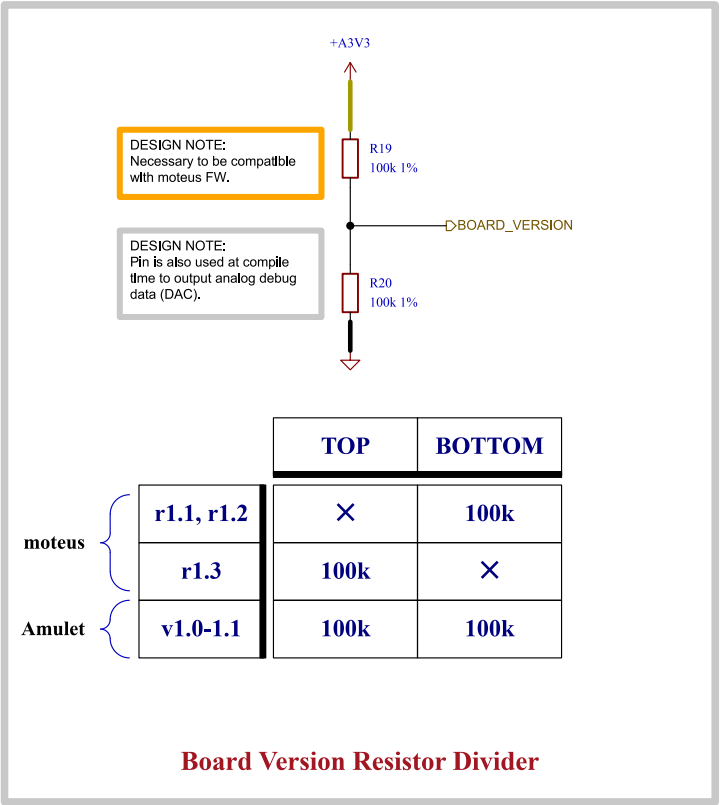
Date:
2024-01-25

Size:
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Revision:
1.1

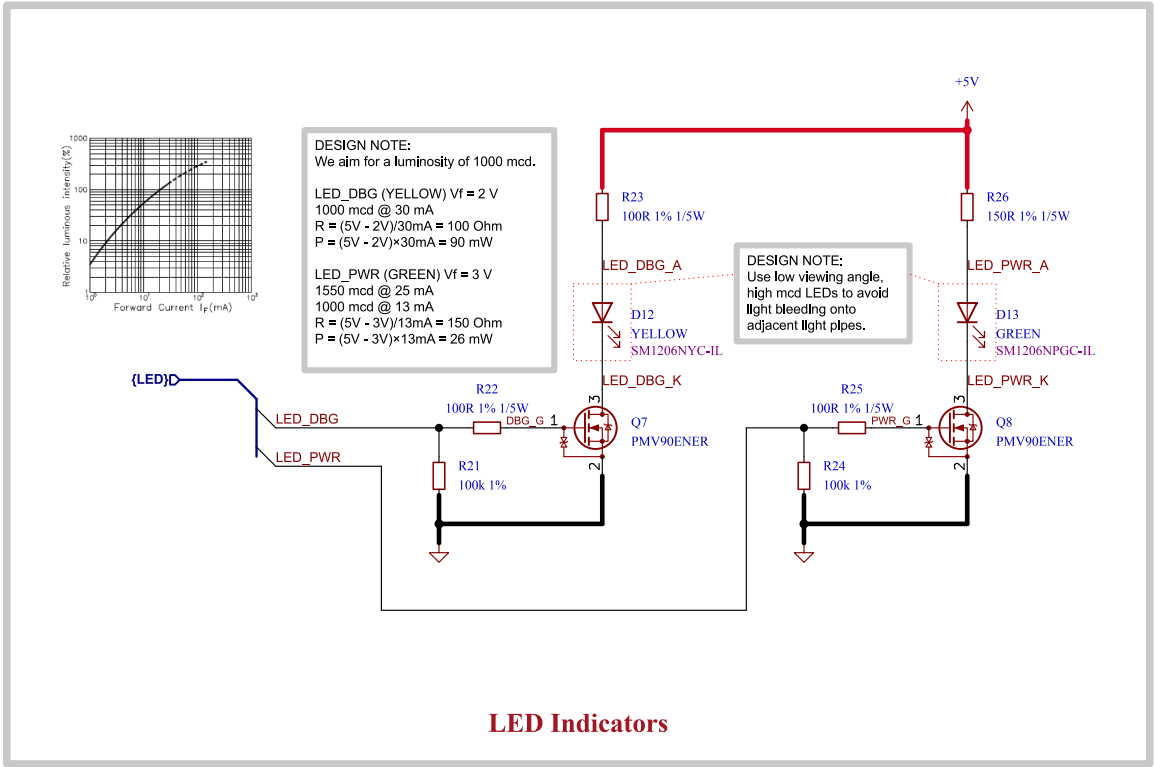
Sheet:
9 of 21

[10] Misc - Board Version, DAC



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			Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Misc - Board Version, DAC		File Name: Misc - Board Version DAC.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
	Sheet Path: /Project Architecture/Misc - Board Version, DAC/			Reviewer:	Size: A4	Sheet: 10 of 21

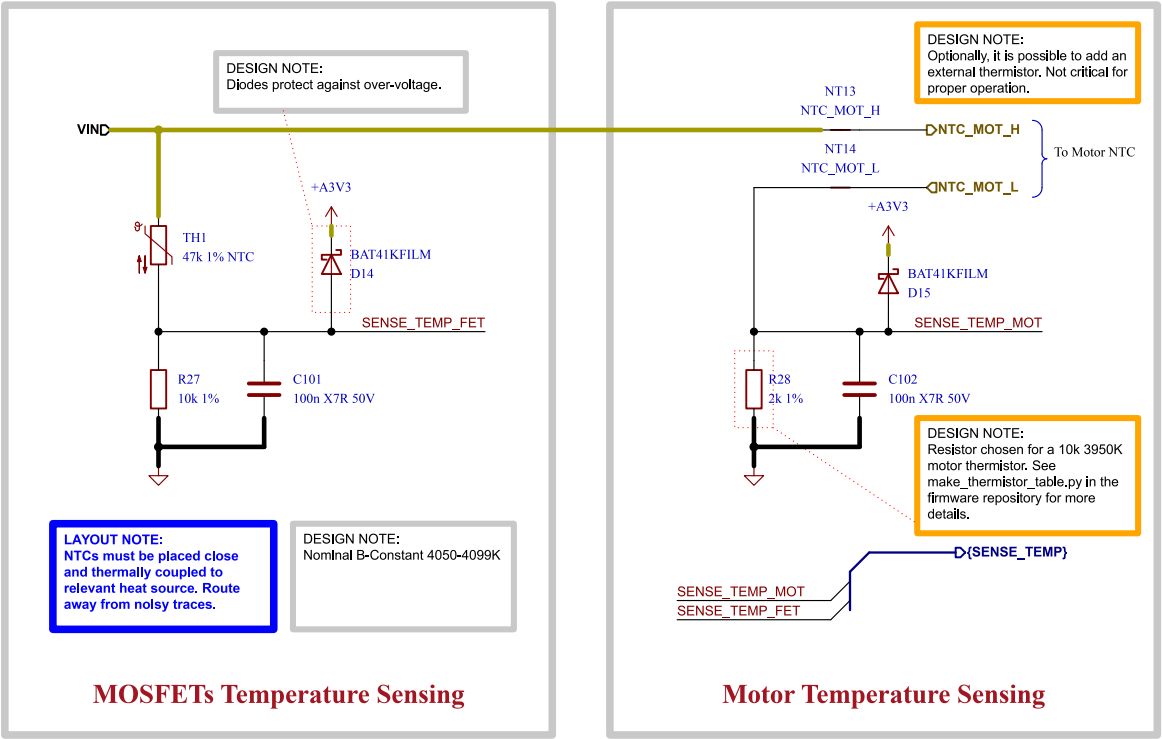
[11] User - LED Indicators



LED Indicators

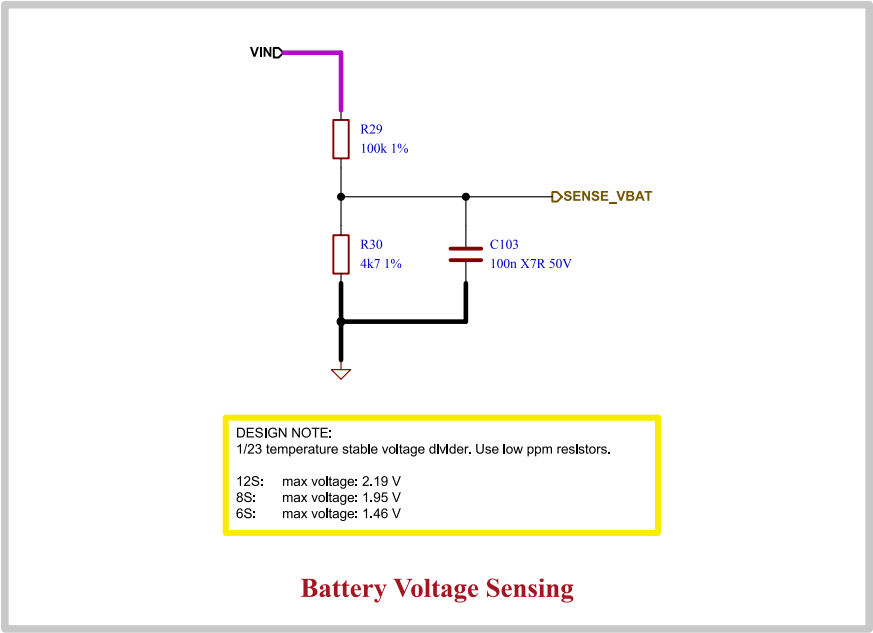
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	Sheet Path: /Project Architecture/User - LED Indicators/		Reviewer:	Size: A4	Sheet: 11 of 21

[12] Sensing - Temperature



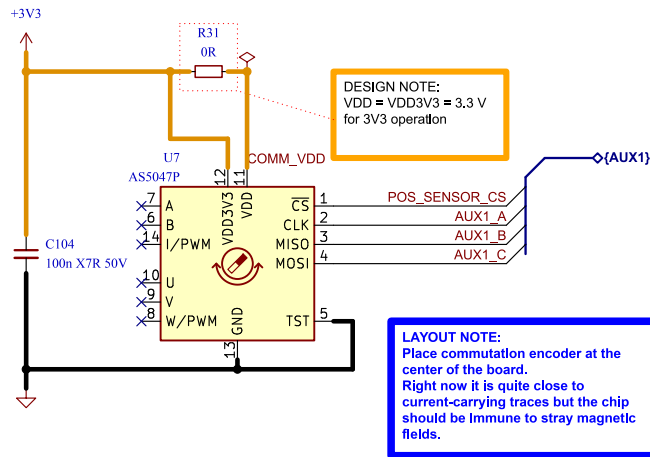
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[13] Sensing - Battery



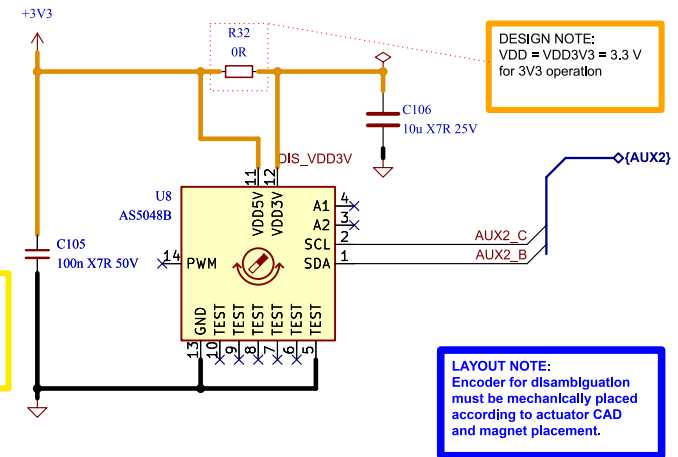
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	Sheet Path: /Project Architecture/Sensing - Battery/			Reviewer:	Size: A4	Sheet: 13 of 21

[14] Sensing - Position



14-Bit On-Axis Magnetic Rotary Position Sensor (commutation)

DESIGN NOTE:
AS5047P senses magnet mounted on planetary sun gear, for commutation.
AS5048B senses magnet mounted on shaft with same reduction factor as planetary gearbox for disambiguation.



14-Bit On-Axis Magnetic Rotary Position Sensor (disambiguation)

	Comments:		Company: EPFL Xplore Research				Variant: Released			
			Board Name: Amulet Motion Controller		Project Name: Chienpanzé					
	Sheet Title: Sensing - Position		File Name: Sensing - Position.kicad_sch		Designer: Vincent Nguyen		Date: 2023-10-14		Revision: 1.1	
Sheet Path: /Project Architecture/Sensing - Position/					Reviewer:		Size: A4		Sheet: 14 of 21	

A

B

C

D

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7

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4

5

[15] Interface - RS-422

DESIGN NOTE:
RS-422 through UART Is used
mostly for absolute position sensors
(e.g. Aksim-2, SSI protocol)

DESIGN NOTE:
100-ohm termination in
Interconnects sheet.

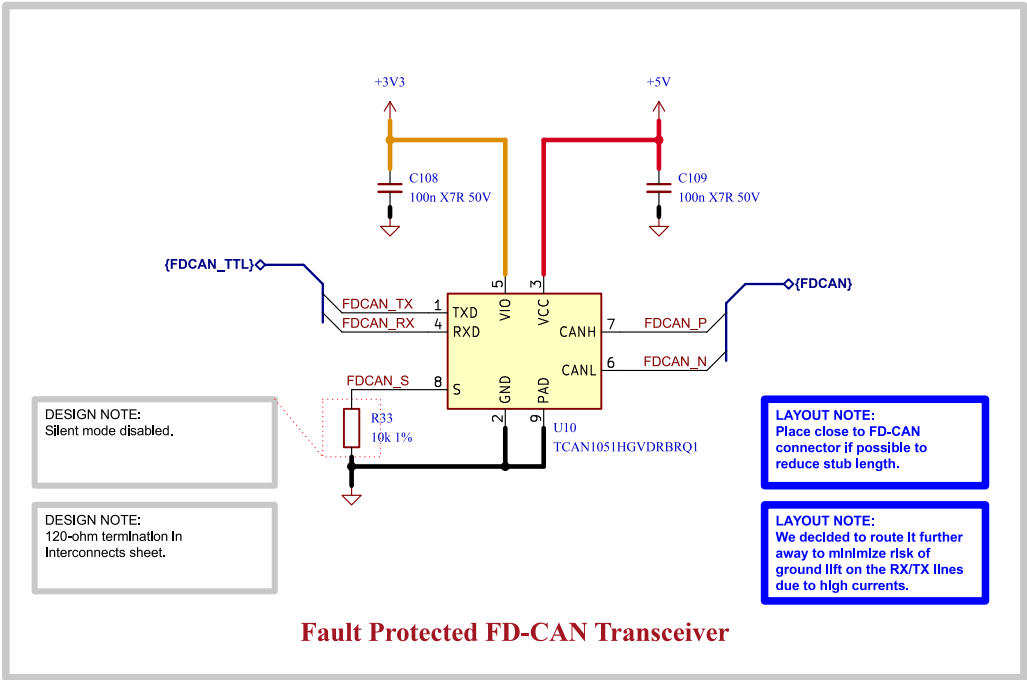
LAYOUT NOTE:
Place close to RS-422
connector.

3.3-V Full-Duplex RS-422 Interface

	Comments:		Company: EPFL Xplore Research		Variant: Released	
					Project Name: Chienpanzé	
	Sheet Title: Interface - RS-422		File Name: Interface - RS-422.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-15	Revision: 1.1
Sheet Path: /Project Architecture/Interface - RS-422/			Reviewer:		Size: A4	Sheet: 15 of 21

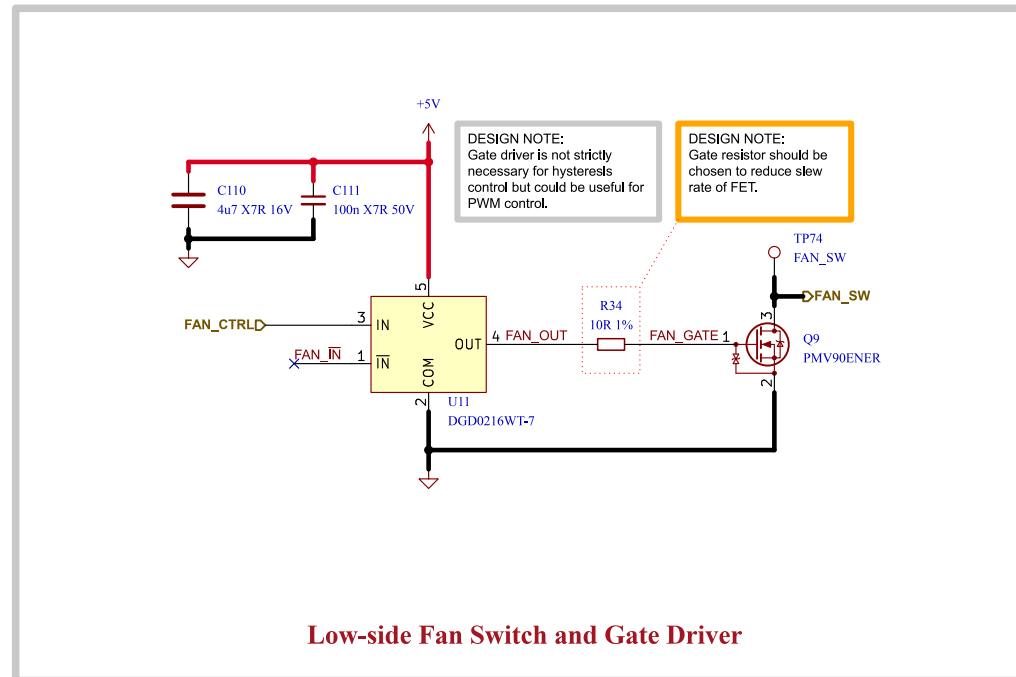
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		Board Name: Amulet Motion Controller			Project Name: Chienpanzé			
	Sheet Title: Interface - RS-422	File Name: Interface - RS-422.kicad_sch	Designer: Vincent Nguyen		Date: 2023-10-15		Revision: 1.1	
	Sheet Path: /Project Architecture/Interface - RS-422/		Reviewer:		Size: A4	Sheet: 15 of 21		

[16] Interface - FD-CAN



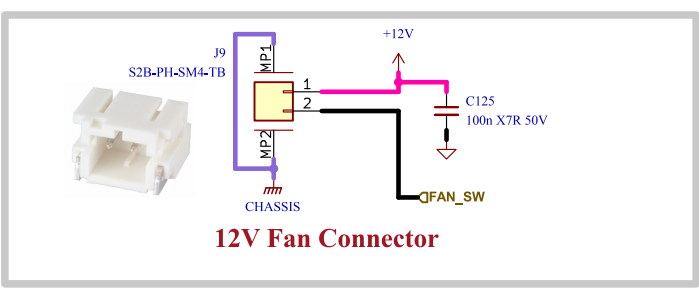
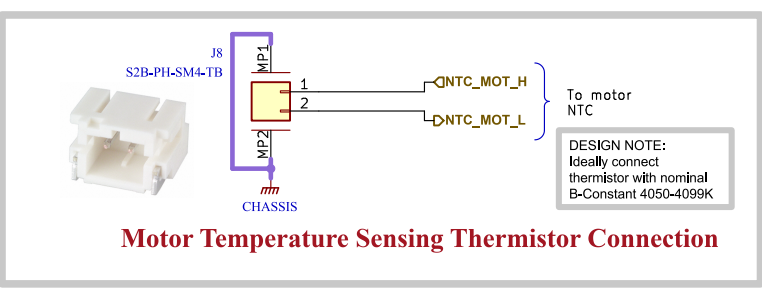
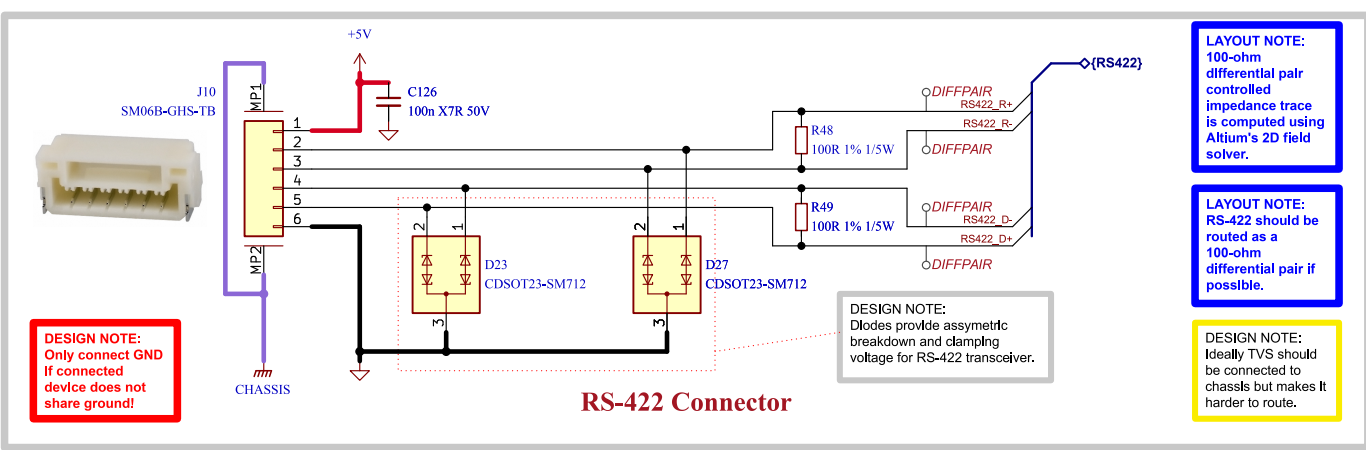
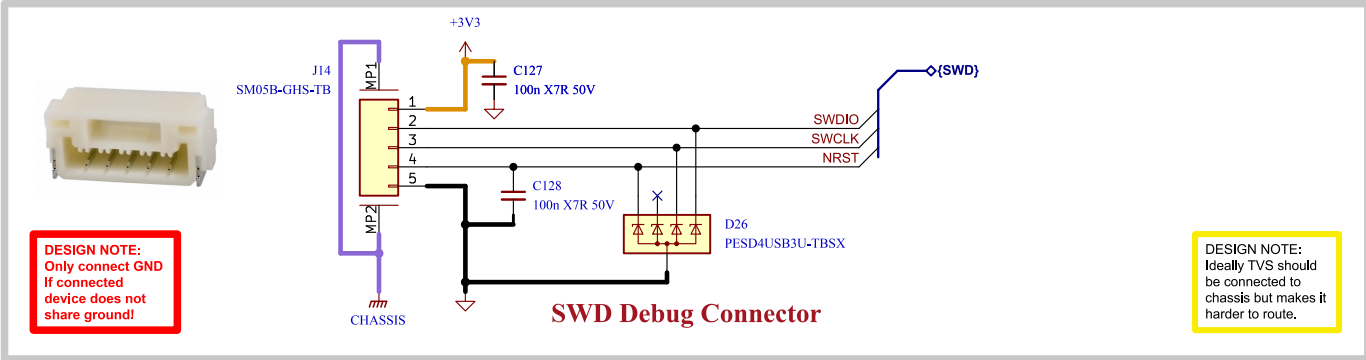
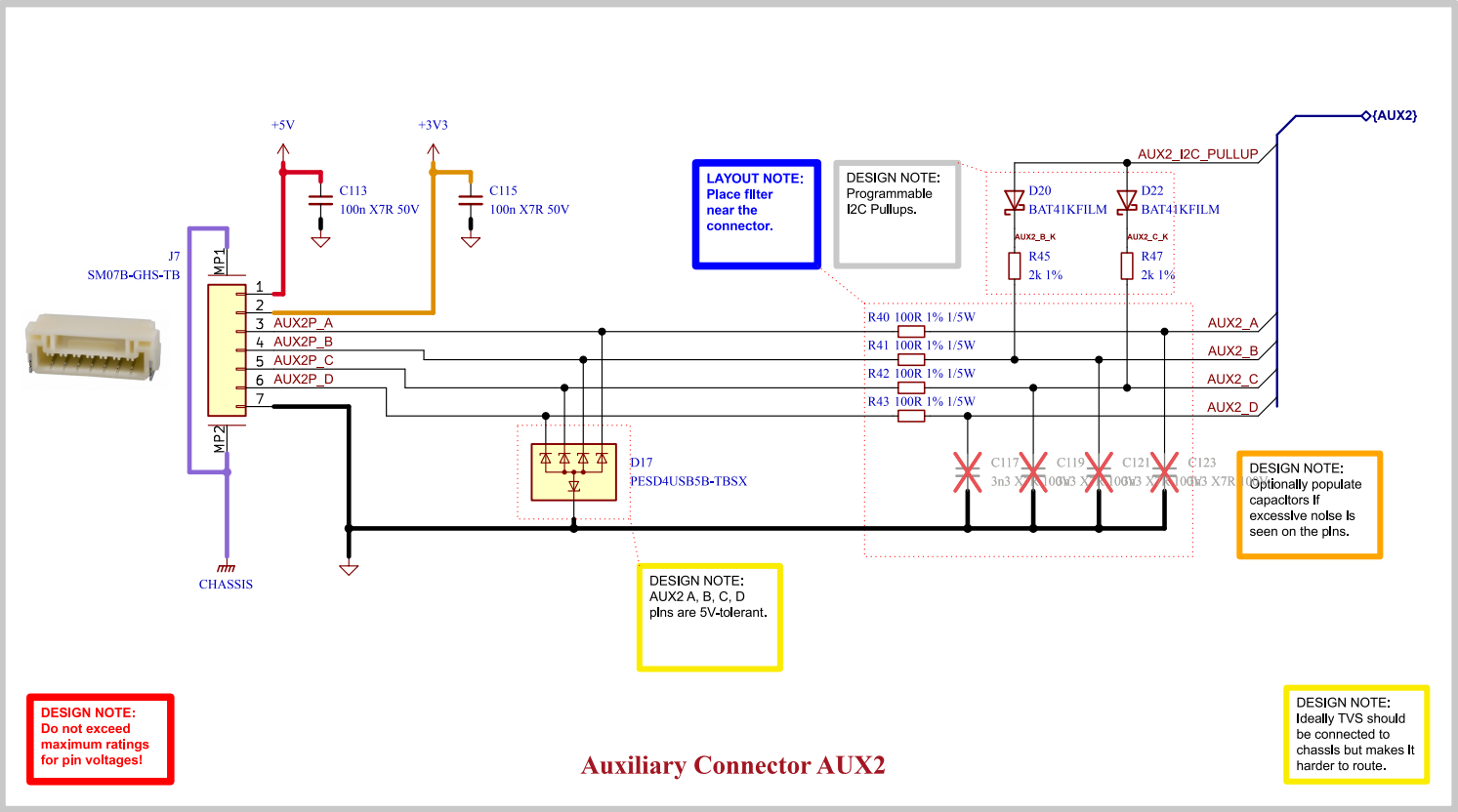
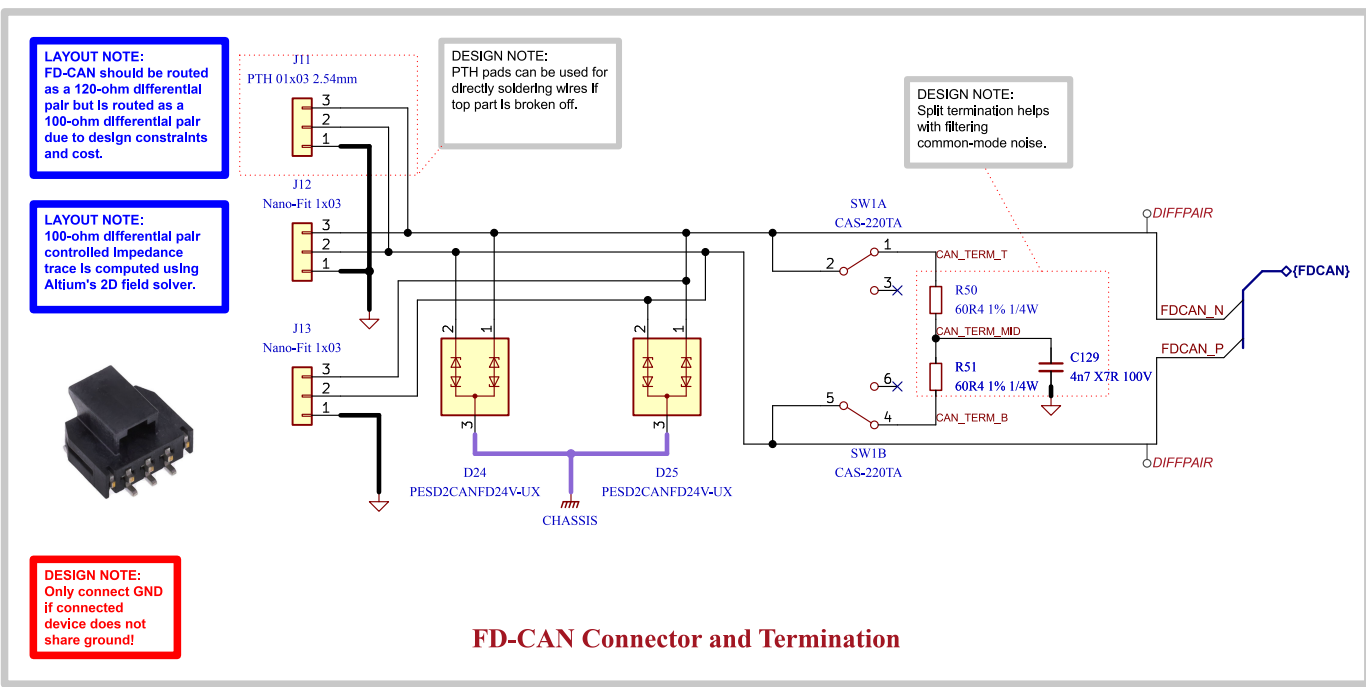
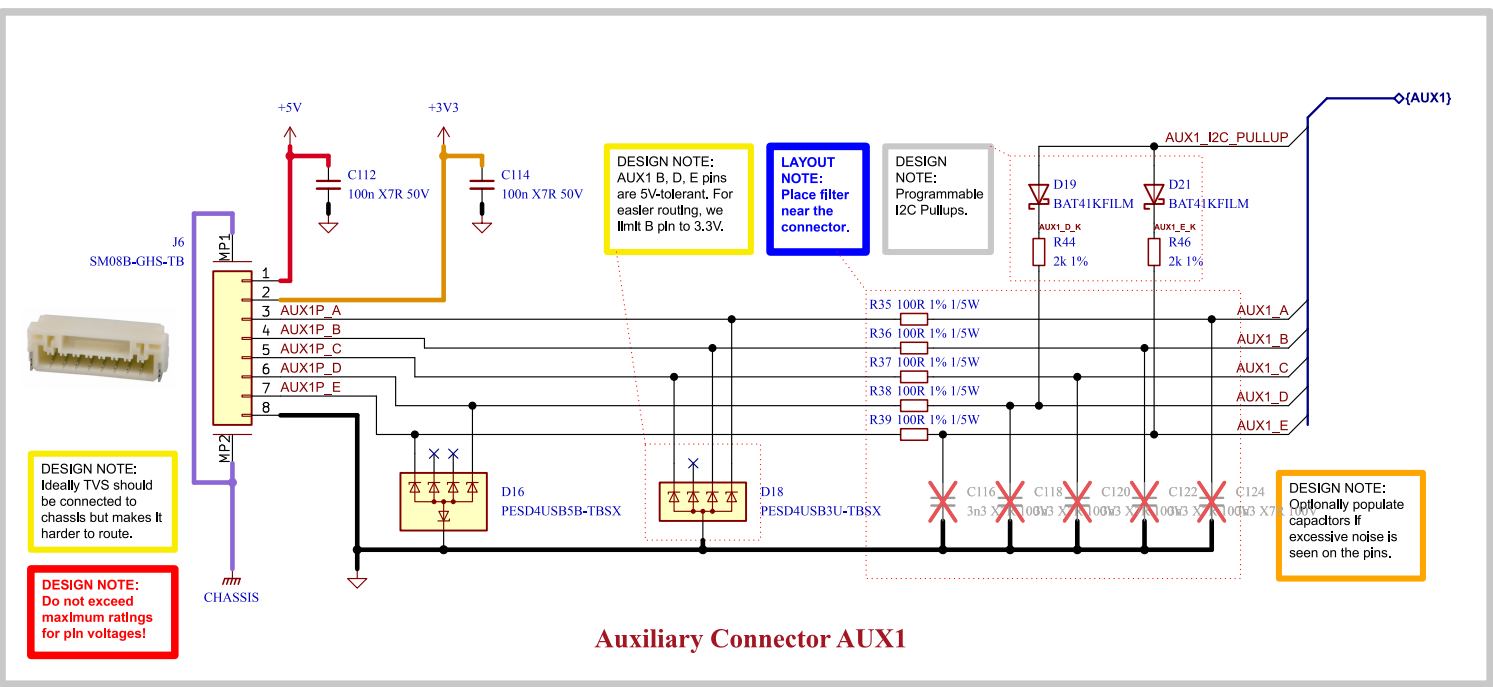
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	Sheet Title: Interface - FD-CAN		File Name: Interface - FD-CAN.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-15	Revision: 1.1
	Sheet Path: /Project Architecture/Interface - FD-CAN/			Reviewer:	Size: A4	Sheet: 16 of 21

[17] Interface - Fan Control



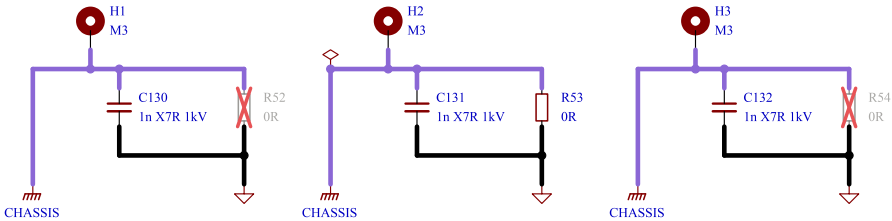
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Sheet Title: Interface - Fan Control	File Name: Interface - Fan Control.kicad_sch	Designer: Vincent Nguyen	Date: 2023-11-19	Revision: 1.1	
Sheet Path: /Project Architecture/Interface - Fan Control/		Reviewer:	Size: A4	Sheet: 17 of 21	

[18] Interface - Interconnects



Comments: Reference: Flexible I/O worked examples	Company: EPFL Xplore Research		Variant: Released	
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - Interconnects	File Name: Interface - Interconnects.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31
Sheet Path: /Project Architecture/Interface - Interconnects/		Reviewer:		Revision: 1.1
			Size: A3	Sheet: 18 of 21

[19] Misc - Holes, Fiducials



DESIGN NOTE:
Optional RC filter for chassis
grounding. Capacitors offer multi-point
low-impedance connections at
high-frequencies.

DESIGN NOTE:
0-Ohm resistor provides low-impedance
bond between chassis and signal GND.
Be aware of ground loops through cable
shield.

Mounting Holes and Chassis Connection

Front



Back



Fiducial Markers

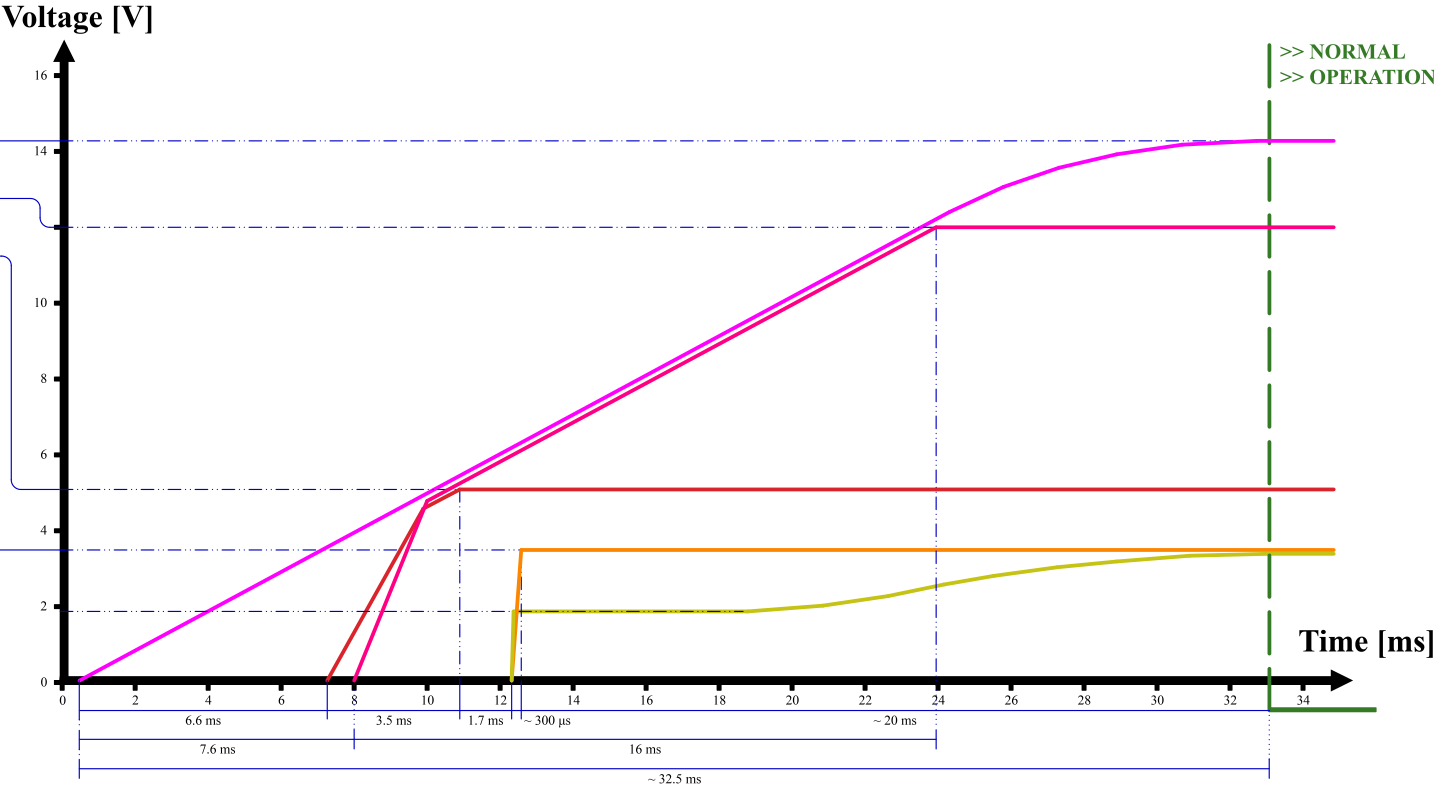
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	Sheet Title: Misc - Holes, Fiducials	File Name: Misc - Holes Fiducials.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-22	Revision: 1.1	
	Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:	Size: A4	Sheet: 19 of 21	

[20] Power - Sequencing

NAME	SOURCE	LEVEL
+VBAT	BATTERY	12 - 44V
+12V	LMR36006	12V $\pm$ 1.5%
+5V	LMR36506	5V $\pm$ 1.5%
+3V3	TPS62172	3.3V $\pm$ 3%
+A3V3	LP2992	3.3V $\pm$ 0.5%

DESIGN NOTE:
Power sequencing graph measured experimentally.

DESIGN NOTE:
Start-up time of 12V rail depends on +VBAT charging time, which itself depends on the precharge circuit.



	Comments:	Company: EPFL Xplore Research	Variant: Released	
		Board Name: Amulet Motion Controller	Project Name: Chienpanzé	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Vincent Nguyen	Date: 2024-03-12
	Sheet Path: /Power - Sequencing/	Reviewer:	Size: A4	Revision: 1.1

[illegible]